

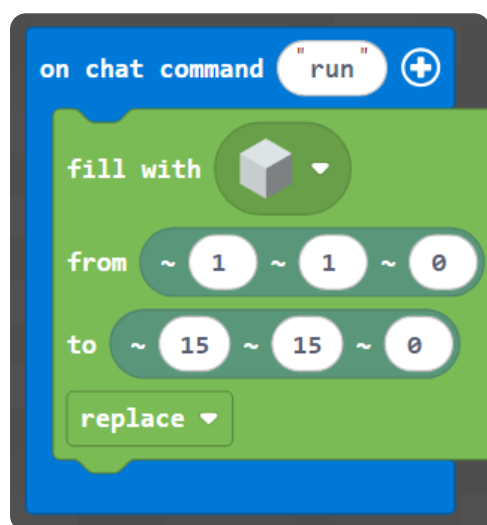
3D Coordinates · Pixel Art 2 — Cat & Tree (x, y) — English Worksheet (Beginner)

Topic: Pixel Art 2 — Cat & Tree (x, y) · **Course:** 3D Coordinates · **Level:** Beginner · **Time:** about 30 minutes

This lesson you copy a **cat** and a **tree**. Each block sits at **(x, y)** — x is **across**, y is **up**.

Build your canvas. Run this to make a blank **15 × 15** wall, then put a **red block at your feet** as your **home spot**:

Blocks



Python

```

def on_run():
    blocks.fill(WHITE_CONCRETE,
                pos(1, 1, 0),
                pos(15, 15, 0),
                FillOperation.REPLACE)
    player.on_chat("run", on_run)
  
```

● **Stand on the red block every time you run.** If you move, the picture moves too.

1 · Predict 🧠

Read the steps. Circle your answer.

```

place brown block at (8, 1)
place brown block at (8, 2)
place brown block at (8, 3)
  
```

Across or up? Circle one: across · up

Which number changes? Circle one: x · y

```

place orange block at (2, 15)
place orange block at (3, 15)
  
```

Across or up? Circle one: across · up

2 · Spot the Bug 🐛

This should make a **tree trunk** going **up** at $x = 8$: (8, 1), (8, 2), (8, 3). But all three blocks land in the **same spot**.

```
place brown block at (8, 1)
place brown block at (8, 1)
place brown block at (8, 1)
```

Hint: to go **up**, the **y** number must grow: 1, 2, 3.

Fill in the blanks to fix it:

```
place brown block at (8, 1)
place brown block at (8, __)
place brown block at (8, __)
```

Which numbers fill the blanks? Circle one: 2 and 3 · 1 and 1

Why was the first code wrong? Circle one: the y never changed · the color was wrong

Stand on your **home spot** (the red block).

Warm up first. Copy this small tree. Read each square's (x, y) and place that color there.

 Small tree

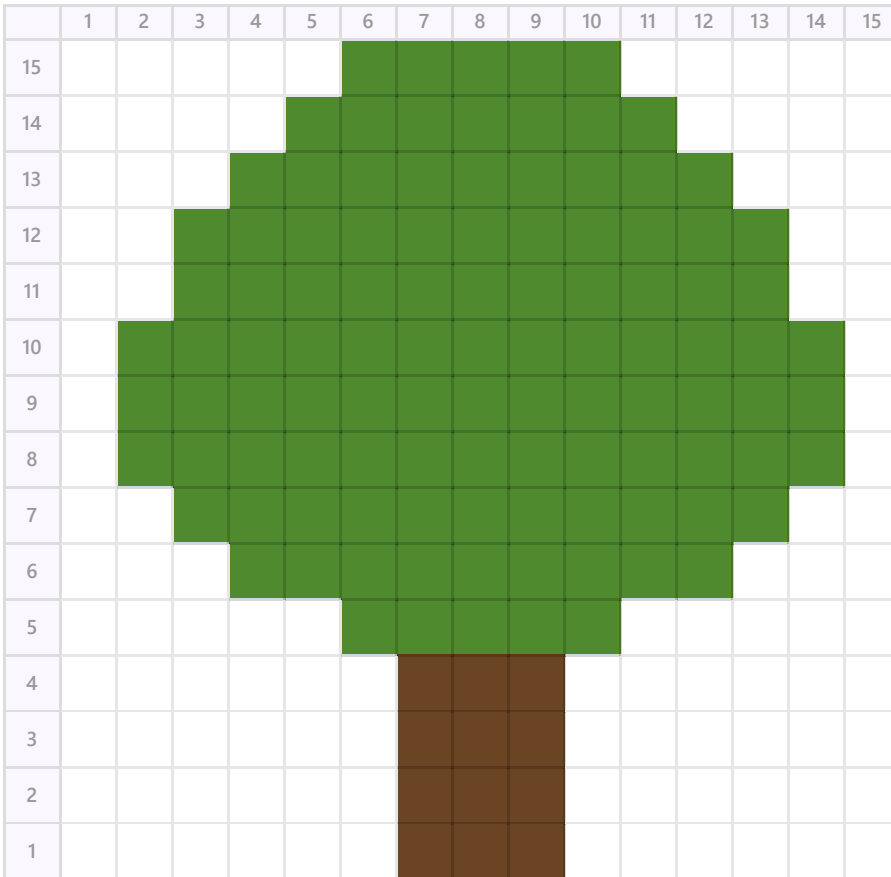
	1	2	3	4	5
5			■		
4		■	■	■	
3			■		
2			■		
1			■		

Try a big one. Pick the 🐱 cat or the 🌳 tree below and copy it the same way. Read each colored square and place that color at its (x, y).

 Cat face

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
15		■	■										■	■	
14		■	■	■								■	■	■	
13		■	■	■	■						■	■	■	■	
12		■	■	■	■	■	■	■	■	■	■	■	■	■	
11		■	■	■	■	■	■	■	■	■	■	■	■	■	
10		■	■	■	■	■	■	■	■	■	■	■	■	■	
9		■	■		■	■				■	■		■	■	
8		■	■	■	■	■	■		■	■	■	■	■	■	
7		■	■	■	■	■		■			■	■	■	■	
6		■	■	■	■	■		■			■	■	■	■	
5		■	■	■	■	■	■	■	■	■	■	■	■	■	
4			■	■	■	■	■	■	■	■	■	■	■	■	
3			■	■	■	■	■	■	■	■	■	■	■	■	
2				■	■	■	■	■	■	■	■	■	■		
1						■	■	■	■	■					

Tree



Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 • Your code. Scroll through it. Say what each part does.

2 • Your build. Point the camera. Name the parts.

Say these out loud, filling in the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.